SOFTWARE TEST PLAN (STP)

Online Exam Portal — SecureExam-AI

AI-Powered Secure Online Examination System with Smart Proctoring and Analytics

Project	Online Exam Portal (SecureExam-AI)
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1. Introduction

Purpose: This document defines the Software Test Plan (STP) for SecureExam-AI (Online Exam Portal) v1.0. It outlines the test objectives, scope, strategy, resources, schedule, deliverables, and responsibilities required to verify that the platform meets the functional and non-functional requirements defined in the project's Software Requirements Specification (SRS).

Project summary: SecureExam-AI is a MERN-stack (MongoDB, Express.js, React + Vite, Node.js) web application that conducts secure, timed online examinations with browser-based AI proctoring (face monitoring, tab-switch and fullscreen-exit detection), automatic evaluation, result publication, and an Admin analytics dashboard with Excel report export. The codebase is organised into three primary modules: frontend (React + Vite + Tailwind), backend (Express REST API + Mongoose/MongoDB, JWT authentication), and ai_proctoring (browser-based monitoring logic).

Scope: Testing covers Student registration and login, Admin exam and question management, the timed fullscreen exam-attempt workflow with auto-submission, AI proctoring (face monitoring, tab/fullscreen violation detection and auto-submit on threshold breach), automatic evaluation and result publication, the Admin analytics dashboard, and Excel-based result export. Live human-invigilator video calling, payment/billing modules, and integration with external university ERP systems are excluded, consistent with the SRS scope statement.

Definitions / Acronyms

- SRS — Software Requirements Specification
- STP — Software Test Plan
- RTM — Requirements Traceability Matrix
- JWT — JSON Web Token
- API — Application Programming Interface
- ODM — Object Document Mapper (Mongoose)
- MCQ — Multiple Choice Question
- RBAC — Role-Based Access Control
- AI — Artificial Intelligence (browser-based proctoring logic)
- TC — Test Case
- UAT — User Acceptance Testing
- WCAG — Web Content Accessibility Guidelines

2. Test Items

The following modules/components of SecureExam-AI are within the scope of this test plan, mapped to the project's actual folder structure:

Test Item	Repository Location	Description
Authentication Module	backend/controllers, routes (auth)	Student/Admin registration, login, JWT issuance, bcrypt password hashing, RBAC middleware
Exam Management Module	backend/controllers, models (Exam)	Admin create/edit/delete exam (title, duration, schedule)
Question Management Module	backend/controllers, models (Question)	Admin add/edit/delete questions with options and correct answers

Test Item	Repository Location	Description
Exam Attempt Module (Student)	frontend/pages, backend routes	Student view of available exams, fullscreen exam attempt UI, live countdown timer, manual/auto submit
AI Proctoring Module	ai_proctoring/	Camera-based face monitoring, tab-switch detection, fullscreen-exit detection, violation counter, auto-submit on threshold
Result & Evaluation Module	backend/controllers (results)	Automatic evaluation on submission, result/history view for students
Admin Analytics Module	frontend (Recharts), backend (/api/admin/stats)	Dashboard totals, performance statistics, score distribution
Excel Export Module	backend (ExcelJS), /api/export/excel/:examId	Generation and download of result reports
Profile Management Module	frontend/pages, backend routes	Student profile view/update

3. Features to be Tested

Features are mapped to SRS requirement IDs (Section 4 & 5 of the SRS) so that every planned test case can be traced back to a documented requirement (see Section 13 — RTM).

3.1 Functional Features

SRS ID	Feature
OEP-F-001	Student registration with name, email, and password
OEP-F-002	Login and JWT issuance for Student/Admin
OEP-F-003	Password hashing with bcryptjs (no plaintext persistence)
OEP-F-004	Role-based access control on Admin-only routes
OEP-F-005	Student profile view and update
OEP-F-010	Admin: create new exam (title, duration, schedule)
OEP-F-011	Admin: edit existing exam details
OEP-F-012	Admin: delete exam (blocked if active attempts exist)
OEP-F-013	Admin: add question(s) with options and correct answer
OEP-F-014	Admin: edit/delete existing questions
OEP-F-015	Admin: list registered students and their attempts
OEP-F-020 .. 025	Student: view available exams; start attempt in fullscreen; live countdown timer; answer navigation; manual submit; auto-submit on time expiry
OEP-F-030 .. 034	AI Proctoring: camera-based face monitoring; tab-switch detection; fullscreen-exit detection; violation counter; auto-submit on violation threshold
OEP-F-040 .. 043	Automatic evaluation of objective answers; result/history publication to student; Admin analytics aggregation; Excel export (name, email, exam, score, %, date)

3.2 Non-Functional Features

SRS ID	Feature	Category
OEP-NF-001	Page load / navigation response ≤ 2s for 90% of requests	Performance
OEP-NF-002	99.5% availability during scheduled exam windows	Reliability
OEP-NF-003	Secure storage of passwords and JWT secrets	Security/Compliance
OEP-NF-004	Retention of attempt and violation logs (audit)	Audit / Data Retention
OEP-NF-005	Usable on 1366×768+ screens; WCAG 2.1 AA colour contrast	Usability/Accessibility

3.3 Security Features

SRS ID	Feature
PRJ-SR-001	TLS 1.2+ mandatory for client-server communication in production
PRJ-SR-002	All passwords hashed (salted, bcryptjs); no plaintext logging/storage
PRJ-SR-003	Short-lived signed JWTs with role claim; expired/tampered tokens rejected
PRJ-SR-004	Question bank correct answers never exposed in student-facing API responses pre-submission
PRJ-SR-005	AI proctoring violation events recorded in an append-only, timestamped audit log per attempt

4. Features Not to be Tested

- Live human-invigilator video calling — not implemented; explicitly out of scope per SRS Section 1.2
- Payment/billing modules — not part of the current version
- Integration with external university ERP systems — out of scope for the current version
- Third-party camera/webcam driver-level behaviour — assumed to work per browser vendor; only the application's use of the MediaDevices API is tested
- MongoDB Atlas / cloud-provider infrastructure internals — assumed reliable; only the application's data access layer is tested
- Underlying npm library internals (React, Express, Mongoose, ExcelJS, Recharts) — assumed tested upstream; only their integration within SecureExam-AI is verified

5. Test Approach / Strategy

5.1 Test Levels

- Unit tests — individual controller/service functions and React components (e.g., password hashing, JWT generation, score calculation, timer countdown logic)
- Integration tests — Frontend ↔ Backend REST API, Backend ↔ MongoDB, AI proctoring module ↔ Exam attempt component
- System tests — End-to-end Student and Admin workflows across the full stack
- Acceptance tests — Project Guide/Evaluator sign-off against SRS acceptance criteria (UAT)

5.2 Test Types

- Functional testing — all Student and Admin features listed in Section 3.1
- Regression testing — re-run of core suites after each defect fix or feature change
- Performance testing — page load/API response time under normal and peak (concurrent exam attempt) load
- Usability & accessibility testing — UI clarity, responsiveness at 1366×768+, colour-contrast spot checks
- Security testing — authentication, authorization, token handling, data exposure (see 5.3)
- Compatibility testing — Chrome, Edge, Firefox (per SRS Section 2.4 operating environment)

5.3 Security Validation

- Verify PIN-equivalent (password) handling — hashing, no plaintext logging (PRJ-SR-002)
- Verify TLS 1.2+ enforcement in production configuration (PRJ-SR-001)
- Verify JWT expiry, signature, and role-claim tampering are rejected (PRJ-SR-003)
- Verify question bank / answer keys are never present in pre-submission student-facing API responses (PRJ-SR-004)
- Verify AI-proctoring violation events are recorded in an append-only, timestamped audit log (PRJ-SR-005)
- Basic input-fuzzing on registration/login fields and exam-answer submission endpoints
- Verify protected/Admin routes return 403 for Student tokens and 401 for missing/invalid tokens

5.4 Entry Criteria

- Stable build deployed to the test environment (frontend + backend + MongoDB instance)
- SRS and this Test Plan reviewed and approved
- Test data (dummy student/admin accounts, sample exams and questions) available
- Test environment configured with required environment variables (PORT, MONGO_URI, JWT_SECRET)

5.5 Exit Criteria

- 100% of planned test cases executed
- 0 open critical/high-severity defects
- All High-priority functional requirements (Section 3.1) verified and passed
- RTM shows full requirement coverage with all mapped test cases passed

6. Test Environment

6.1 Hardware

- Desktop/laptop with a functioning webcam (required for AI proctoring tests)
- Stable broadband internet connection
- Minimum display resolution 1366×768 (per OEP-NF-005)

6.2 Software

- Frontend: React + Vite build served via npm run dev / production build
- Backend: Node.js runtime running Express.js REST API (npm start)
- Database: MongoDB (local instance or MongoDB Atlas sandbox)

- Browsers: Chrome, Edge, Firefox (latest evergreen versions) with camera permission enabled

6.3 Tools

Tool	Purpose
Postman	REST API testing (auth, exams, questions, results, admin, export endpoints)
Jest / Mocha + Supertest	Backend unit and integration testing
React Testing Library	Frontend component unit testing
Cypress / Selenium WebDriver	End-to-end UI automation (Student & Admin workflows)
JMeter	Performance/load testing of REST APIs
MongoDB Compass	Database inspection (verify hashed passwords, audit logs, no plaintext data)
GitHub Issues / Jira	Defect and test-case tracking
Lighthouse / axe DevTools	Usability and accessibility (WCAG 2.1 AA) checks

6.4 Test Data

- Dummy Student accounts (multiple, for concurrent-attempt and RBAC testing)
- Dummy Admin account(s)
- Sample exams with a mix of question counts and durations
- Sample question banks with valid, boundary, and invalid option/answer combinations

7. Test Schedule

Indicative milestones (to be finalised with the course/project timeline):

Milestone	Target Date
Test plan review & sign-off	21-Sep-2026
Test case design (all modules)	22-Sep-2026 to 25-Sep-2026
Test environment setup	26-Sep-2026
Test execution — Cycle 1 (functional, security, AI proctoring)	25-Sep-2026 to 02-Oct-2026
Defect fixing & retest — Cycle 2 (regression)	03-Oct-2026 to 07-Oct-2026
Performance & usability/accessibility testing	08-Oct-2026 to 10-Oct-2026
UAT with Project Guide/Evaluator	12-Oct-2026 to 13-Oct-2026
Test Summary Report & sign-off	14-Oct-2026

8. Test Deliverables

- Software Test Plan (this document)
- Detailed Test Cases (manual and automated) — see Appendix A
- Automated test scripts (Jest/Supertest, Cypress/Selenium)
- Test data sets
- Test execution logs

- Defect reports
- Requirements Traceability Matrix (Section 13)
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
Project Guide / Evaluator	Sheela Devi	Review test plan, approve results, sign-off readiness
QA Lead	Korada Sannath Kumar	Prepare and maintain the test plan, coordinate execution
Test Engineer	Kondapuram Charith Sai Reddy	Design and execute test cases, log and track defects
Developer	DEERAJ S	Support defect triage and fixes
Student Team (Project Owner)	Krishna H Zalavadiya	Provide build, environment, and requirement clarifications

Note: team member names to be filled in by the student team.

10. Risks and Mitigation

Risk	Mitigation
Camera/webcam permission denied or unavailable on a test machine, blocking AI proctoring tests	Maintain a dedicated test machine with a verified webcam; document a fallback/mock camera stream for CI
Delay in stable build delivery from development	Request early smoke-tested builds; run smoke tests before full cycles begin
Test environment (MongoDB/API) downtime	Maintain a local MongoDB fallback instance alongside Atlas sandbox
False positives/negatives in tab-switch or fullscreen-exit detection across browsers	Test explicitly on Chrome, Edge, and Firefox; log browser-specific behaviour as known issues
Flaky end-to-end tests due to timer-based auto-submit logic	Use configurable/shortened timers in the test environment and explicit waits in automation scripts
Limited team bandwidth (academic project)	Prioritise High-priority requirements first; defer Low-priority/nice-to-have test cases if time-constrained

11. Assumptions & Dependencies

- The MongoDB instance (local or Atlas sandbox) will be stable and available throughout testing
- Test accounts (Student/Admin) and sample exam data will be seeded before execution begins
- Test machines have functioning webcams and up-to-date evergreen browsers
- Backend environment variables (PORT, MONGO_URI, JWT_SECRET) are correctly configured
- No external ERP, payment, or live-invigilation integrations are required, consistent with SRS scope

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment (frontend, backend, or database) is unavailable for more than 4 hours
- The build is unstable enough to block more than 30% of planned test cases
- A critical security defect (e.g., answer keys exposed pre-submission, broken RBAC) is found — testing of the affected module is suspended pending a fix

Resume testing if:

- Blocking defects are resolved and verified in a new build
- The test environment is confirmed stable

13. Test Case Management & Traceability

The Requirements Traceability Matrix (RTM) below maps each SRS requirement to its responsible module and reference test case(s). Status uses N (Not run) / P (Pass) / F (Fail), to be updated during execution.

Req ID	Requirement (short)	SRS Section	Module	Test Case(s)	Status
OEP-F-001	Student registration	4.1	AuthModule	TC-Auth-01	N
OEP-F-002	Login & JWT issue	4.1 / DS-Auth-01	AuthModule	TC-Auth-02	N
OEP-F-003	Password hashing (bcryptjs)	4.1	AuthModule	TC-Auth-03	N
OEP-F-004	RBAC on Admin routes	4.1	AuthModule	TC-Auth-04	N
OEP-F-005	Student profile update	4.1	AuthModule	TC-Auth-05	N
OEP-F-010	Create exam	4.2 / DS-Exam-01	ExamModule	TC-Exam-01	N
OEP-F-011	Edit exam	4.2	ExamModule	TC-Exam-02	N
OEP-F-012	Delete exam	4.2	ExamModule	TC-Exam-03	N
OEP-F-013	Add question	4.2 / DS-Exam-02	QuestionModule	TC-Q-01	N
OEP-F-014	Edit / delete question	4.2	QuestionModule	TC-Q-02	N
OEP-F-015	List students & attempts	4.2	AdminModule	TC-Admin-01	N
OEP-F-020..022	View exams; start fullscreen; live timer	4.3	AttemptModule	TC-Attempt-01..03	N
OEP-F-023..024	Answer navigation; manual submit	4.3	AttemptModule	TC-Attempt-04..05	N
OEP-F-025	Auto-submit on time expiry	4.3	AttemptModule	TC-Attempt-06	N
OEP-F-030	Camera face monitoring	4.4 / DS-Proctor-01	ProctorModule	TC-Proct-01	N
OEP-F-031	Tab-switch detection	4.4	ProctorModule	TC-Proct-02	N
OEP-F-032	Fullscreen-exit detection	4.4	ProctorModule	TC-Proct-03	N

Req ID	Requirement (short)	SRS Section	Module	Test Case(s)	Status
OEP-F-033	Violation counter	4.4	ProctorModule	TC-Proct-04	N
OEP-F-034	Auto-submit on violation threshold	4.4	ProctorModule	TC-Proct-05	N
OEP-F-040	Auto evaluation & result	4.5 / DS-Result-01	ResultModule	TC-Result-01	N
OEP-F-041	Result history to student	4.5	ResultModule	TC-Result-02	N
OEP-F-042	Admin analytics dashboard	4.5	AnalyticsModule	TC-Analytics-01	N
OEP-F-043	Excel export	4.5	ExportModule	TC-Export-01	N
OEP-NF-001	Response time target	5 / DS-Perf-01	WebUI / API	TC-Perf-01	N
OEP-NF-002	99.5% availability	5	WebUI / API	TC-Ops-01	N
OEP-NF-003	Secure storage (passwords/JWT)	5	AuthModule	TC-Sec-01	N
OEP-NF-004	Log retention (1 academic year)	5	ProctorModule / DB	TC-Ops-02	N
OEP-NF-005	Usability & WCAG 2.1 AA	5	WebUI	TC-UX-01	N
PRJ-SR-001	TLS 1.2+ enforced	5.1.2	Infra / API	TC-Sec-02	N
PRJ-SR-002	Password hashing	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-03	N
PRJ-SR-003	JWT expiry/tamper rejection	5.1.2	AuthModule	TC-Sec-04	N
PRJ-SR-004	No answer-key exposure pre-submission	5.1.2	QuestionModule / API	TC-Sec-05	N
PRJ-SR-005	Append-only violation audit log	5.1.2	ProctorModule	TC-Sec-06	N

14. Test Metrics & Reporting

Metrics collected

- % test cases executed
- % passed / failed
- Defect density (defects per module)
- Defect aging (time open)
- Requirement coverage (from RTM)

Reports

- Daily/periodic execution status
- Final Test Summary Report

15. System Models — UML Use-Case Diagrams

The following diagrams (as provided) define the actor–use-case scope for Student and Admin, and directly inform the test items and features in Sections 2 and 3.

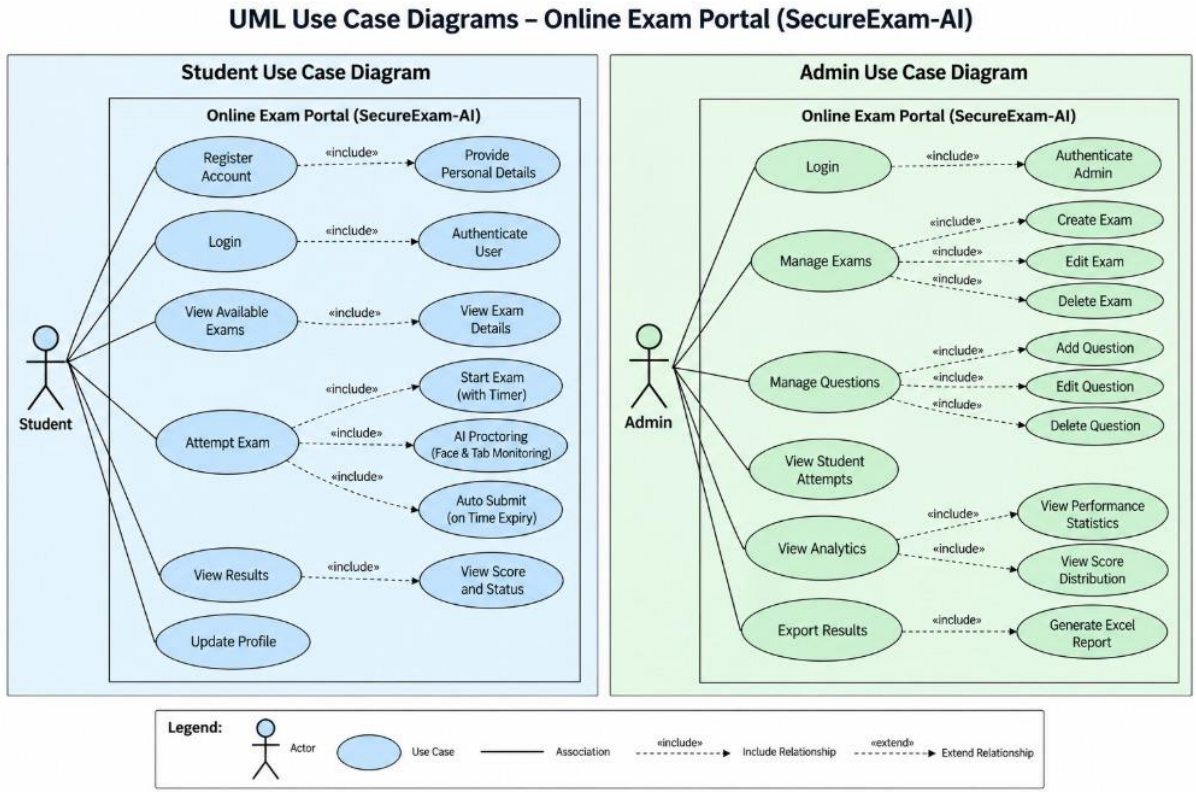


Figure 1: UML Use-Case Diagrams — Online Exam Portal (SecureExam-AI). Left: Student use cases (Register, Login, View Available Exams, Attempt Exam, View Results, Update Profile). Right: Admin use cases (Login, Manage Exams, Manage Questions, View Student Attempts, View Analytics, Export Results).

Appendix A — Sample Detailed Test Cases

A representative subset of detailed test cases is provided below, covering the highest-priority requirements across all modules. The full test-case set should be expanded from this template to reach 100% RTM coverage (Section 13) before execution begins.

TC ID	Description	Pre-conditions	Steps	Expected Result	Priority
TC-Auth-01	Valid student registration	App running; email not already registered	1. Open Register page 2. Enter name, unique email, valid password 3. Submit	Account is created; user is redirected to Login	H
TC-Auth-02	Login with valid credentials	Registered account exists	1. Open Login page 2. Enter correct email/password 3. Submit	A valid signed JWT is returned; user lands on role-appropriate dashboard	H
TC-Auth-02b	Login with invalid credentials	Registered account exists	1. Enter wrong password 2. Submit	Login is rejected with an appropriate error; no token issued	H
TC-Auth-03	Password stored hashed	A registration has been completed	1. Inspect the users collection in MongoDB Compass	Password field contains only a bcrypt hash, never plaintext	H
TC-Auth-04	RBAC blocks student from Admin route	Valid student JWT	1. Call an Admin-only endpoint (e.g., create exam) using a student token	API responds 403 Forbidden; no data is modified	H
TC-Exam-01	Admin creates a new exam	Logged in as Admin	1. Open Manage Exams 2. Enter title, duration, schedule 3. Save	Exam is created and appears in the exam list	H
TC-Exam-03	Delete exam blocked when active attempts exist	Exam has at least one in-progress attempt	1. Attempt to delete the exam as Admin	Deletion is blocked with a clear error; exam remains listed	M
TC-Q-01	Admin adds a question with options and correct answer	Logged in as Admin; an exam exists	1. Open the exam's question manager 2. Add question text, options, correct answer 3. Save	Question is stored and linked to the exam	H
TC-Attempt-01	Student starts exam in fullscreen	Student logged in; exam available in window	1. Select exam 2. Click Start	Browser enters fullscreen mode; countdown timer starts; camera permission is requested	H
TC-Attempt-06	Exam auto-submits on time expiry	Exam attempt in progress	1. Let the countdown timer reach 00:00 without manual submission	Attempt is automatically submitted; student	H

TC ID	Description	Pre-conditions	Steps	Expected Result	Priority
				is redirected to results/confirmation	
TC-Proct-02	Tab-switch is detected and counted	Exam attempt in progress, fullscreen active	1. Switch to another browser tab during the attempt	A violation is logged and the on-screen violation counter increments by 1	H
TC-Proct-03	Fullscreen-exit is detected	Exam attempt in progress	1. Press Esc / exit fullscreen during the attempt	A violation is logged; student is prompted to re-enter fullscreen	H
TC-Proct-05	Auto-submit on violation threshold	Exam attempt in progress	1. Repeat tab-switch/fullscreen-exit until the configured violation threshold is reached	Exam is automatically submitted; violation events are recorded in the audit log	H
TC-Result-01	Objective answers auto-evaluated on submit	Exam attempt submitted	1. Submit an exam with known correct/incorrect answers	Score matches the expected count of correct answers	H
TC-Analytics-01	Admin analytics dashboard reflects data	At least one exam with submitted attempts exists	1. Open Admin → Analytics	Totals, performance statistics, and score distribution match underlying attempt data	M
TC-Export-01	Excel export contains correct fields	At least one exam with results exists	1. Admin exports results for an exam via /api/export/excel/:examId	Downloaded .xlsx contains student name, email, exam, score, percentage, submission date for every attempt	M
TC-Sec-04	Expired/tampered JWT is rejected	A previously valid JWT is available	1. Wait for token expiry OR modify the token payload 2. Call a protected endpoint	API responds 401 Unauthorized; access is denied	H
TC-Sec-05	Answer keys not exposed before submission	Exam attempt in progress	1. Inspect API responses returned to the student client while answering questions	Correct-answer fields are absent from all pre-submission responses	H
TC-Perf-01	Page load / navigation performance	Application deployed to test environment	1. Load exam list and question navigation under normal load using JMeter	90th percentile response time ≤ 2 seconds	M
TC-UX-01	Usability & colour-contrast spot check	Application deployed	1. Run Lighthouse/axe on Student and Admin dashboards at 1366×768	No critical accessibility violations; contrast	L

TC ID	Description	Pre-conditions	Steps	Expected Result	Priority
				meets WCAG 2.1 AA where applicable	

16. Approvals

Role	Name	Signature / Date
Project Guide / Evaluator	Sheela Devi	
QA Lead	Korada Sannath Kumar	16/9/2026
Product owner	Krishna H Zalavadiya	16/9/2026
Developer	DEERAJ S	16/9/2026
Test Engineer	Kondapuram Charith Sai Reddy	16/9/2026